

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm: 

FIRST SOLVE
New here? Do the walkthrough once at [cubepath-six.vercel.app/learn](#). This card is the memory jog.
NOTATION
R U F L D B = turn that face clockwise, looking at it from outside the cube.
= counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.
WORDS
algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit
1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm — work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS
Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.
**white sticker RIGHT**
RUR'U'
**white sticker FRONT**
L'ULU
**white sticker UP** run the first one once to knock it out, then look again
RUR'U'
3 MIDDLE EDGES
Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.
**goes RIGHT**
U RUR'U' y L'ULU
**goes LEFT**
U' L'ULU y' RUR'U'
Edge stuck in the wrong slot? Run either one to kick it out, then place it.



 **RUR'U'** sexy move

 **RUR'U** Sune

 **R'FRF'** sledgehammer

 gap = change grip

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TWO-LOOK
OLL CROSS · yellow edges
**Line bar left-to-right**
FRUR'U'F'
**Hook hook pointing front-right**
FRUR'U'F'
OLL CORNERS · yellow face
**Sune**  **RUR'U RU2R'**
**Anti-Sune**  **RU2R' U'RUR'**
**Pi**  **FRUR'U'F' FRUR'U'F'**
No yellow edge at all? Run the first one, then read again.

 **RUR'U'** sexy move

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 **R'FRF'** sledgehammer

 gap = change grip

OLL CORNERS · continued
**Headlights**  **R2DR'U2 RD'R'U2R'**
**2x Headlights**  **FRUR'U'F' URU2R'**
**Chameleon**  **rUR'U'r' FRF'**
**Bowtie**  **F'FRUR'U'r' FR**
FALLBACK
The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; three is the worst.
Top not all yellow and nothing matches? Run Sune and read again.



TWO-LOOK
OLL CROSS · yellow edges
**Line bar left-to-right**
FRUR'U'F'
**Hook hook pointing front-right**
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The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked; three is the worst.
Top not all yellow and nothing matches? Run Sune and read again.



ONE-LOOK PLL
ADJACENT CORNER SWAP · exactly one face shows headlights
**T L headlights: pairs B-left F-left**
RUR'U' R'F R2U'R'U' RUR'F'
**Ja L solid: pairs B-right F-left R-front**
x R2FRF' RU2 r'Ur U2
**Jb L solid: pairs B-left F-right R-back**
RUR'F' RUR'U' R'F R2U'R'
**Aa L headlights: pairs F-right R-front**
x L2D2 L'UL D2 L'UL'
**Ab L headlights: pairs B-right R-back**
x' L2D2 LUL' D2 LU'L
**F L solid**
R'U'F' RUR'U' R'F R2U'R'U' RUR'UR
One face shows headlights: two corners of that face match. Hold as drawn, run, then turn the top to finish.

 **RUR'U'** sexy move

 **RUR'U** Sune

 **R'FRF'** sledgehammer

 gap = change grip

ADJACENT CORNER SWAP · continued
**Ra L headlights: pairs F-left**
RUR'U' RUR DR'URD' R'U2R'
**Rb L headlights: pairs B-left**
R2F RURU'R' F'R U2R'U2R
**Ga L headlights: pairs F-right**
R2UR'UR'U' RU'R2 U'D R'URD'
**Gb L headlights: pairs B-back**
R'UR' UD' R2UR'URU' RU'R2D
**Ge L headlights: pairs B-right**
R2U'R'URU' RU'R2 UD' RU'R'D
**Gd L headlights: pairs R-front**
RUR' U'D R2U'R'UR' UR'UR2D'
Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day — one is the mirror of the other.



ONE-LOOK PLL
ADJACENT CORNER SWAP · exactly one face shows headlights
**T L headlights: pairs B-left F-left**
RUR'U' R'F R2U'R'U' RUR'F'
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RUR'F' RUR'U' R'F R2U'R'
**Aa L headlights: pairs F-right R-front**
x L2D2 L'UL D2 L'UL'
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 **RUR'U'** sexy move

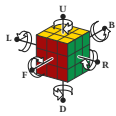
 **RUR'U** Sune

 **R'FRF'** sledgehammer

 gap = change grip

ADJACENT CORNER SWAP · continued
**Ra L headlights: pairs F-left**
RUR'U' RUR DR'URD' R'U2R'
**Rb L headlights: pairs B-left**
R2F RURU'R' F'R U2R'U2R
**Ga L headlights: pairs F-right**
R2UR'UR'U' RU'R2 U'D R'URD'
**Gb L headlights: pairs B-back**
R'UR' UD' R2UR'URU' RU'R2D
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Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day — one is the mirror of the other.



ANNEX — BIG CUBES AND THE MAP
BIG CUBES · 4x4 and 5x5
Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY — never widen it.
last 2 edges
4x4 PLL parity
4x4 OLL parity · 1 edge pair flipped · half the time
Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'
Rw U2 x Rw U2 Rw U2 3Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'
Fix parity during the last layer, before OLL and PLL — both parity algorithms move corners.
NOTATION

Each letter turns that face clockwise, seen from outside the cube — so from your seat L, D and B look counter-clockwise. = counter-clockwise, 2 = half turn.
M = middle slice, turning the way L turns. x y z = the whole cube, turning the way R, U and F turn. Lowercase r f = that face plus the layer behind it, turning together.
WORDS
algorithm = a fixed sequence of turns · trigger = a short algorithm your hands run as one unit · sexy move = the R U F' U' trigger · sledgehammer = the R' F R' F' trigger · parity = a case a 3x3 cannot have · edge pair = two edges glued into one, on a 4x4 or 5x5
 **RUR'U'** sexy move

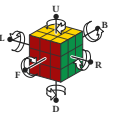
 **RUR'U** Sune

 **R'FRF'** sledgehammer

 gap = change grip

BEGINNER SHORTCUT · use it until you know Sune
Twist a yellow corner four turns at a time, turning ONLY the top face between corners.
yellow faces RIGHT
R'DRD x2
yellow faces FRONT
D'RD x2



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Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd LAYER ONLY — never widen it.
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Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'
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R'DRD x2
yellow faces FRONT
D'RD x2



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exactly 80 mm: _____

FAST SOLVE

4 YELLOW CROSS

yellow edges only — ignore the corners



hook **no yellow edge** at all
FRUR'U'

hook **two at a right angle**: hold them **BACK** and **LEFT**
FRUR'U'

hook **line** hold the bar **left-to-right**
FRUR'U'

One algorithm, up to three times: dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn U so at least two top edges match the center under them. The first seven moves are Sune, used on every card after this one, here Sune gets one U on the end.



two matched, side by side hold them **BACK** and **hook**
RU'R'U' RU2R' U

two matched, opposite run it once from any hold, then hook and **hook**
RU'R'U' RU2R' U

6 PUT THE YELLOW CORNERS HOME



one corner home hold it **FRONT-LEFT**
RU'L'U' RU'L'

no corner home run it once from any hold, then look again
RU'L'U' RU'L'

This cycles the other three corners into place. It also twists the yellow face you just built — expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'DRD x2

yellow faces FRONT
D'RDR x2

Between corners ONLY the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look destroyed while you do this. Keep going — to the last top turn brings it back.

4 FIRST SOLVE

4 YELLOW CROSS - yellow edges only – ignore the corners



Don't do yellow edge at all
FRUR'U'F



Hook two at a right angle: hold them **BACK** and **LEFT**
FRUR'U'F



Line hold the bar left-to-right
FRUR'U'F

One algorithm, up to three times; don't hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn U so at least two top edges match the center under them. The first seven moves are a line, used on every card after this one; here we sume Gets U on the end.



two matched, side by side hold them **BACK** and **RIGHT**
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6 PUT THE YELLOW CORNERS HOME



one corner hold it **FRONT-LEFT**
RU'L'U' RU'L



no corner **home** run it once from any hold, then look
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This cycles the other three corners into place. It also twists the yellow face you just built – expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS

yellow faces RIGHT
R'D'R'D x2



yellow faces FRONT
R'D'R'D x2

Between corners turn **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look destroyed while you do this. Keep going – the last top turn brings it back.

PLL CORNERS headlights = two matching corners on one face



T headlights on ONE face – hold that face **LEFT**
RUR'U' R'F RZUR'U' RUF'

Y no headlights – the two swapping corners sit diagonally
FRU'R'U' RUR'F RUR'U' R'RF'

PLL EDGES – corners home, top still not solved



Ub the front edge travels **LEFT**
R2U RUR'U' R'U' RUR'



Uu the front edge travels **RIGHT**
M2U MU2 MUM2

H both pairs swap straight across
M2U'M2 U2 M2U'M2



Z each pair swaps with its neighbour
M'U'R'U2M2U' M'U2 M2U

START HERE – two new algorithms finish any last layer
 Yellow cross: **F R U R' U'** until the cross appears – at most 3.
 Yellow face: **Sune**, read again, repeat – at most 3 (see the badges).
 Corners: **T-Ferm**. No headlights to hold left! Run it twice.
 Edges: **Ub**. No solved edge to put at the back! Run it twice.
 Each case on the front replaces one repeat with one run. Three a week; you can solve the whole time.

H V S Z
 Both show headlights on all four faces. The edge between them decides: it belongs to the opposite face = **H**, to a neighbour = **Z**.
STUCK?
 Corners home but nothing solves? Turn the top face once and read again – that is often the whole fix.
 One case has beaten you five times? Park it, use its fallback above, come back next week.
 A single piece looks twisted and no algorithm touches it? The cube was assembled wrong. Put it out and reset it.

WORDS
OLL = make the whole top face yellow - **PLL** = slide the top pieces home - headlights = two matching corners on one face - **AUF** = the free top turn that lines a case up

PLL CORNERS

headlights = two matching corners on one face



T headlights on ONE face – hold that face **LEFT**
RUR'U' R'F R'F' RUR'U' R'UR'



Y no headlights – the two swapping corners sit diagonally
FR'U'R' UR'U'R' RUR'U' R'FR'

PLL EDGES – corners home, top still not solved



Ub the front edge travels LEFT
R2U RUR'U' R'U' R'UR'



Ua the front edge travels RIGHT
M2U MU2 MUM2



H both pairs swap straight across
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Z each pair swaps with its neighbour
M2UM2M2U' M2U M2U

START HERE

two new algorithms finish any last layer
 Yellow cross: **F R U R' U'** until the cross appears – at most 3.
 Yellow face: Same, read again, repeat – at most 3 (see the badges).
 Corners: Turn them. No headlights to hold left! Run it twice.
 Edges: Ub on the front edge to put at the back? Run it twice.
 Each case on the front replaces one repeat with one run. Three a case you can solve the whole time.

H VS Z

Both swap headlights on all four faces. The edge between them decides it: belongs to the opposite face = H, to a neighbour = Z.

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Corners home but nothing solves? Turn the top face once and read again – that is often the whole fix.
 One case has beaten you five times! Park it, use its fallback above, come back next week.

A single piece looked twisted and no algorithm touches it? The cube was scrambled wrong. Pop it out and reset it.

WORDS

OIL = make the whole top face yellow – PLL = slide the top pieces home – headlights = two matching corners on one face – AUF = the free top turn that's a case up

EDGES ONLY

corners already home



- U or L holds, F, L, R headlights; 3 edges counter-clockwise

M2U M2U M'U M2



- U or L holds, F, R headlights; 3 edges clockwise

R2U R'U' R' U' R'U' R'



- H all 4 headlights; opposite edges swap

M2U'M2 U2 M2U'M2



- M'U all 4 headlights; adjacent edges swap

M'U M2U'M2U' M'U2 M2U

DIAGONAL CORNER SWAP

no headlights any where



- Y pairs F-left R-back

FR'U'R' R'U'R' R'U'R' R'FR'



- Y pairs F-left L-front

R'U'R' Y R'F' R2U'R'U' R'FR'



- Na pairs B-left F-right L-front R-back

R2U'R R'U'R' R'U'R' R'F' R2U'R' U2R'U'R'



- Nb pairs B-right F-left B-back R-front

R'U'R' R'U'R' F'U'F' L'U'F' R'FR'



- X' undoes matches

X' L'U'LD' L'U'LD' L'U'LD' L'U'LD

STUCK?

Case you have not learned? Two look-in from Card 2 – 2-Perm the 1B still solves it. None of these can strand you.

Beaten five times by one case? Park it, two-look it, come back next week.

EDGE-LOCK

EDGES ONLY corners already home

E = U4 solid, F1 R headlights; 3 edges counter-clockwise

M2U M2U M'U'M2

U = U4 solid, F1 R headlights; 3 edges clockwise

R2U R'U'R' R'U' R'U'R'

H all 4 headlights; opposite edges swap

M2M2 U2 M2U'M2

M' all 4 headlights; adjacent edges swap

M'U' M'U'M2M' M'U' M'U' M2U

DIAGONAL CORNER SWAP

no headlights anywhere

Y pairs F-left R-back

F'R'U'R' R'U'R' R'U'R' R'F'R'

V pairs F-left L-front

R'U'R'U' Y R'F' R'U'R' R'F'R'

Na pairs B-left F-right L-front R-back

R'U'R' R'U'R' R'U'R' R'F'

R2U'R' U2R'U'R'

Nb pairs B-right F-left R-back B-front

R' UR'U'R' F'U'F' R'U'R' F'R'F'

R'U'R'

E nothing matches

F' L' U'L'D' L'U'L'D' L'U'L'D' L'U'L'D'

HOW TO LOOK

- Count the headlight faces. One = front of that card. Four = this column. None = the column on the right.
- Only then read the edges to pick the exact case.
- Line the case up as drawn; run, turn the top again to finish – that last time turns the AUF.

= already yours from Card 2.

WORDS

PL = slide the top pieces home - headlights = two matching corners on one face - **AUF** = the free top turn that lines a case up - adjacent corner swap - the swapping corners share an edge - diagonal corner swap - the swapping corners sit opposite

Card 2 gives you 1972 last layers in one look. This card gives 7172.

Next, the other face (see below) solved as pairs – drilled with a cube in the app: cubuhelm3x3x3 solved as pairs

DONE = all 2020 names created; twenty-one in a row, no separate days, on two separate days. @MASTER_PL:LL 3/4 solves under 0.30. My

THE LADDER
1/4 FIRST SOLVE - 9 unlocks - five solves, each one solve
2/5 TWO-LOCK - +13 - unlock: fifteen last letters in exactly two solves
3/6 ONE-LOCK - +15 - done: all 21 named, twice on separate days
4/7 FULL - full QLL, PZL, cross planning, look-ahead - in the app, not on cards
5/8 THERE ARE ONLY THREE
A card is good at a closed set of cases you tell apart by sight. If it's not, it's bad. There are 36 cases, 36 possible pairs, 36 possible pairs with no finite case list. Full QLL is 57 cases, 22 of which differ by one letter. The 22 are the 22 pairs that are nullified by the app, with a real cube and randomised setup. This set stops where the medium stops.
PRINT AND VERSION
EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were to reverse the build, it should fail more than 50% of the time. Recognition wording is generated from the same priming the diagram is drawn from.
Set V=0 - reprint any single card at cubepals@six-verged.app/print - print at 100% / Actual size, never 'Fit to page'.

THE LADDER

1/4 FIRST SOLVE - 9 cuberhythms - unlock five solves, each face
2/5 TWO-LOOK - +13 - unlock: fifteen last layers in exactly two
3/6 ONE-LOOK PL - +15 - done: all 21 named, twice on separate
days, not on consecutive days

There is a card that, full OLL, F2L, cross planning, look-ahead - in the app, not on cards.

THERE ARE ONLY THREE

A card is good at a closed set of cards you tell apart by sight. It is not good at a wide range of cards you tell apart by feel. The cards with no finite case list. Full OLL is 47 cases, 22 of which differ only by a single cube turn. A wall of 47 cards, each with a different case, with the app, with a real cube and randomised setup. This set stops where the blind cube starts.

PRINT AND VERSION

VERIFIED ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time, if you were to print the cards. If you want to know how the cards were built, see Set v1.0 - reprint any single card at cubepath3-six-vercel-app/print - print at 100% / Actual size, never fit to page.

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The number